Lesson 9.3 - Student's T-Distribution

> **Video:** Student's T Distribution - Confidence Intervals & Margin of Error by The Organic Chemistry Tutor
> **Chapter:** 9 - Confidence Intervals

When to use t instead of z

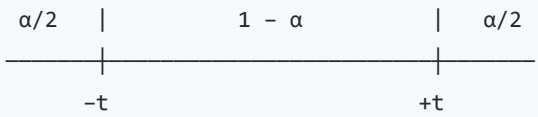
Use the **normal (z)** distribution when...	Use **Student's t** when...
the **population** SD σ is known	only the **sample** SD s is known
$n \geq 30$	$n < 30$

Having only s and a small n are both signs that you need t.

Formulas

$$\bar{x} \pm t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}, \qquad \text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}$$

Symbol	Meaning
$\nu = n - 1$	**Degrees of freedom** (Greek "nu")
$\alpha = 1 - \text{CL}$	The total area outside the interval, i.e. both tails
$\alpha/2 = \dfrac{1 - \text{CL}}{2}$	The **one-tail** area, used to look up t



Reading a t-table: go to the row for ν and the column for the one-tail area $\alpha/2$.

Example 1 - student weights, 95% CI

20 students: $\bar{x} = 165$ lb, $s = 4.5$ lb.

s is a **sample** SD (the school has far more than 20 students) and $n = 20 < 30$, so use t.

- $\nu = n - 1 = 20 - 1 = 19$
- $\alpha/2 = (1 - \text{CL})/2 = (1 - 0.95)/2 = 0.025$
- Table: $t_{19,\,0.025} = 2.093$

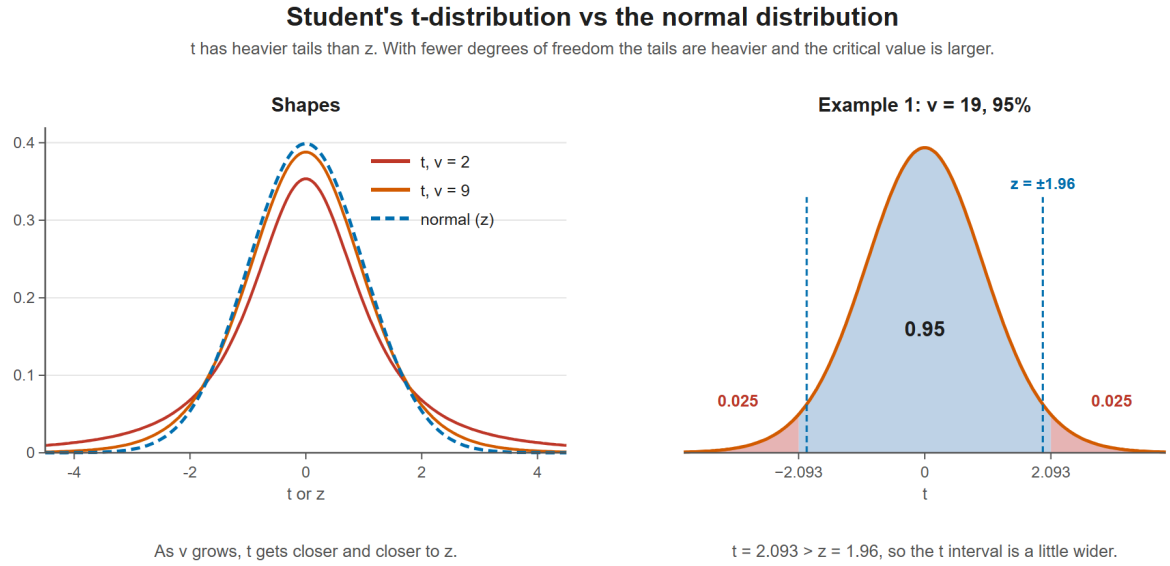
(b) Margin of error

$$\text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}} = 2.093 \times \frac{4.5}{\sqrt{20}} = \frac{9.4185}{4.4721} \approx 2.106$$

(a) Interval

$$\bar{x} \pm \text{EBM} = 165 \pm 2.106 \;\Rightarrow\; 162.894 < \mu < 167.106$$

We are 95% confident that the mean weight of all the school's students is between about **162.9 and 167.1 lb**.



Left: t has heavier tails than the normal curve, especially with few degrees of freedom. Right: for $\nu = 19$ the 95% critical value is 2.093, a bit further out than z = 1.96, so t intervals are slightly wider.

Example 2 - chemistry scores, 90% CI

The class has 500 students. 10 scores were sampled: 76, 84, 69, 92, 58, 89, 73, 97, 85, 77.

(a) Sample mean and SD

$$\bar{x} = \frac{\sum x}{n} = \frac{800}{10} = 80$$

Distances from 80: 4, 4, 11, 12, 22, 9, 7, 17, 5, 3 (squaring removes the signs).

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} = \sqrt{\frac{4^2 + 4^2 + 11^2 + 12^2 + 22^2 + 9^2 + 7^2 + 17^2 + 5^2 + 3^2}{10 - 1}} = \sqrt{\frac{1234}{9}} \approx 11.709$$

(b) Margin of error at 90%

- $\nu = n - 1 = 10 - 1 = 9$
- $\alpha/2 = (1 - \text{CL})/2 = (1 - 0.90)/2 = 0.05$
- Table: $t_{9,\,0.05} = 1.8331$

$$\text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}} = 1.8331 \times \frac{11.709}{\sqrt{10}} \approx 6.787$$

(c) 90% confidence interval

$$\bar{x} \pm \text{EBM} = 80 \pm 6.787 \;\Rightarrow\; 73.213 < \mu < 86.787$$

Key takeaways

- Use t when σ is unknown (you only have s), especially when $n < 30$.
- Degrees of freedom $\nu = n - 1$. Look up the one-tail area $\alpha/2 = (1 - \text{CL})/2$.

- CI = $\bar{x} \pm t\, s/\sqrt{n}$. The t value is larger than z for small samples, which gives a wider and more cautious interval.